Jonnathan Singh Alvarado, Ph.D.

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Education

Postdoctoral Fellow, Neurobiology, Harvard University — 2021–Present

Ph.D., Neurobiology, Duke University — 2014–2021

B.S., Microbiology, University of Florida — 2009–2013

Research Experience

Postdoctoral Fellow, Beth Israel Deaconess Medical Center / Harvard Medical School — 2021–Present
Dr. Mark Andermann

Graduate Student, Duke University — 2014–2021
Dr. Richard Mooney, Department of Neurobiology

Research Assistant, Duke University — 2013–2014
Dr. Cagla Eroglu, Department of Cell Biology

Undergraduate Researcher, University of Florida — 2012–2013
Dr. Stephanie Karst, Department of Virology

Undergraduate Researcher, University of Florida — 2011–2012
Dr. Brent Reynolds, Department of Cancer Biology

Publications

Slow-timescale regulation of dopamine release governs recovery of a motivational drive over days — 2026
Sunkavalli PS, Madara JC, **Singh Alvarado J**, Ventura BF, Baker JG, Zhu P, Christenson LF, Diaz A, Chen J, Lazen E, Ravotto L, Patriarchi T, Andermann ML, Zhang SX. *Neuron* (*in press*)

EASI-PASS: An accessible pipeline for linking functional imaging and mRNA profiling — 2026
Singh Alvarado J, Massengill C, Stern J, Amsalem O, Ventura B, Jang A, Cook S, Veliche A, Sunkavalli P, Patel D, Colaccino J, Evans K, Wang Y, Andermann ML. *bioRxiv* (*preprint*)

The role of modern neuroscience in the renaissance of psychedelic research — 2025
Evans KE, Massengill C, **Singh Alvarado J**. *Revista Costarricense de Psicología*

A sensitive soma-localized red fluorescent calcium indicator for in vivo imaging of neuronal populations at single-cell resolution — 2025
Zhou S, Zhu Q, Eom M, Fang S, Subach OM, Ran C, **Singh Alvarado J**, Sunkavalli PS, Dong Y, Wang Y, Hu J, Zhang H, Wang Z, Sun X, Yang T, Mu Y, Yoon YG, Guo ZV, Subach FV, Piatkevich KD. *PLoS Biology*

Transient cAMP production drives rapid and sustained spiking in brainstem parabrachial neurons to suppress feeding 2024
Singh Alvarado J*, Lutas A*, Madara JC, Isaac J, Lommer C, Massengill C, Andermann ML. *Neuron*

Divergent projections from locus coeruleus to the cortico-basal ganglia system and ventral tegmental area of the zebra finch 2022
Singh Alvarado J*, Hatfield J*, Mooney R. *Journal of Comparative Neurology*

Neural dynamics underlying birdsong practice and performance 2021
Singh Alvarado J, Goffinet J, Michael V, Liberti W, Hatfield J, Gardner T, Pearson J, Mooney R. *Nature*

Building Opportunities and Overtures in Science and Technology: Establishing an early intervention, multi-level, continuous STEM pathway program 2021
Garcia J, Gibson R, Breslav A, **Singh Alvarado J**, Seas A, Maduka A, et al. *Journal of STEM Outreach*

MIN1PIPE: A miniscope 1-photon-based calcium imaging signal extraction pipeline 2018
Lu J, Li C, **Singh Alvarado J**, Zhou ZC, Fröhlich F, Mooney R, Wang F. *Cell Reports*

Expressing mutant huntingtin in the songbird basal ganglia disrupts cortico-basal ganglia networks and vocal sequences 2016
Tanaka M, **Singh Alvarado J**, Murugan M, Mooney R. *Proc. Natl. Acad. Sci. USA*

Astrocytes refine cortical connectivity at dendritic spines 2014
Risher WC, Patel S, Kim IH, Uezu A, Pilaz LJ, **Singh Alvarado J**, Calhan OY, Silver DL, Stevens B, Soderling SH, Eroglu C. *eLife*

Rapid Golgi analysis method for efficient and unbiased classification of dendritic spines 2014
Risher WC, Ustunkaya T, **Singh Alvarado J**, Eroglu C. *PLoS ONE*

* *Equal contribution*

Talks

Special Seminar – UC San Diego School of Biological Sciences	2026
Special Seminar – The Francis Crick Institute	2026
Special Seminar – Interdisciplinary Bioscience DTP, University of Oxford	2026
Joint Endocrinology Lab Meeting – BIDMC	2022, 2024, 2026
Outstanding Postdoc Talk – HMS Neurobiology Departmental Retreat	2026
Optogenetic Approaches to Understanding Neural Circuits and Behavior – Lucca, Italy	2026
Systems Neuroscience Club – Harvard Medical School	2026
Departmental Pizza Talks – Harvard Medical School Department of Neurobiology	2025
Department Lab Results Seminar – Boston Children's Hospital Kirby Neurobiology Center	2024

Special Seminar – University of Pittsburgh & Carnegie Mellon University 2024
SfN Minisymposium – Peripheral Mechanisms of Energy & Metabolism 2023
Special Seminar – Champalimaud Centre for the Unknown 2023
Joint Lab Meeting – Carnegie Mellon University Systems Club 2022
COSYNE (6% of abstracts selected for talks) 2021
Special Seminar – Columbia University 2021
Virtual Dopamine Conference (ViDA) 2021
University of Washington & Allen Institute Neuromodulation Club 2020
Neuromodulation Gordon Research Conference 2019
Duke University Brown Bag Seminars 2018
Duke Institute for Brain Sciences Symposium 2017
Dept. of Cell & Molecular Biology Seminar 2017
Duke Distinguished Donor Weekend 2017
TEDx Duke 2016

Fellowships and Awards

Brain & Behavior Research Foundation (BBRF) Young Investigator Grant 2026–2028
David Hubel Outstanding Postdoc Award – HMS Department of Neurobiology 2026
The Transmitter – Rising Stars of Neuroscience 2025
NINDS K99/R00 Award (parent mechanism), deferred 2025
MOSAIC NINDS K99/R00 Award 2024
Harvard Brain Initiative Postdoctoral Pioneer Award 2024–2025
Harvard Mind, Brain, and Behavior (MBB) Postdoctoral Fellowship 2022–2024
Graduate Student Training Enhancement Grant (GSTEG) 2019
Simons Foundation Neuromodulation GRC – Outstanding Talk Award 2019
Cold Spring Harbor Laboratory – Neural Data Science Course 2019
Duke Institute for Brain Sciences Symposium – Best Talk 2017
Ruth K. Broad Research Fellowship 2016
Biosciences Collaborative for Research Engagement (BioCoRe) Scholar 2015
Latin American & Caribbean Scholarship (4-year scholarship) 2009–2013

Leadership and Service

Gordon Research Seminar Co-chair 2025–Present
Organizer for the 2027 GRS Modulation of Neural Circuits and Behavior, Barcelona.

HPREP Mentor 2024–Present
Lecturer and hands-on project coordinator for the HPREP STEM outreach program at Harvard.

NeuroLaunchpad 2021–2022
Created and secured funding for a trainee-centered nationwide seminar series giving exposure to burgeoning scientists; over 1,000 unique viewers.

BOOST Coach 2014–2021
Developed year-long STEM research programs and hands-on curricula for low-income middle and high school students.

Committee for Reimagining Doctoral Education at Duke 2018–2020
One of three Duke graduate students selected to serve. Met monthly with a group of 30 faculty members, Duke's provost, and the Dean of research training to construct a strategic plan for restructuring and improving doctoral education at Duke.

Other Service, Duke University 2014–2021
Neurobiology Diversity and Inclusion Taskforce, Neurobiology Steering Committee, Dean's Mentoring Award Committee, Gordon G. Hammes Mentoring Award Committee, Faculty Advancement Committee, Alfred P. Sloan Scholars Committee.

Rex Scholar 2016–2017
Helped develop software and virtual lectures distributed widely in North Carolina public schools.

TECHO – Costa Rica NGO 2007–2011
Directed volunteer teams building emergency transitional homes for families in need.